

Mitt første dokument

Arnstein

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse
2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts —
tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(huxtable)
```

Attaching package: 'huxtable'

The following object is masked from 'package:dplyr':

add_rownames

The following object is masked from 'package:ggplot2':

theme_grey

```
options(
  paged.print = FALSE,
  pillar.print_max = 25,
  pillar.print_min = 25
)
```

Quarto

Min egen tekst.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

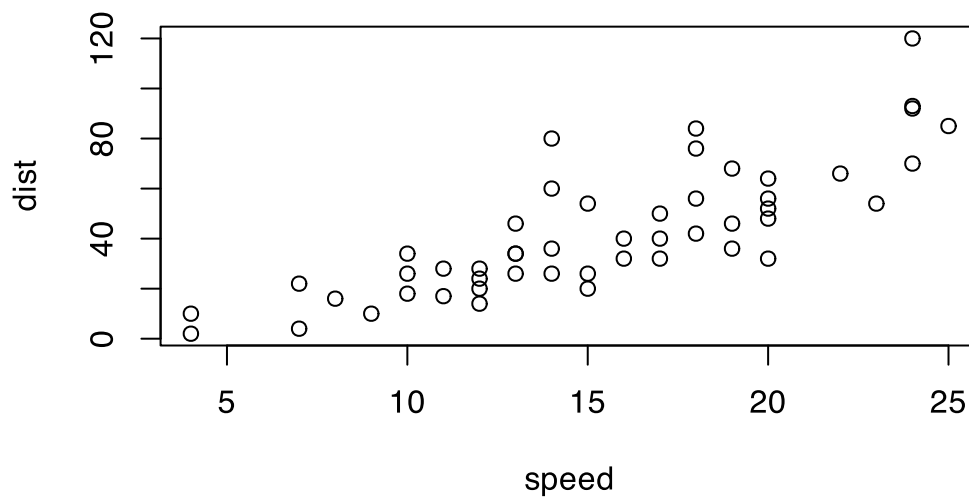
You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

1. første
2. andre
 - c. punkt
 - d. fffhghg
3. tredje

```
plot(cars)
```



```
w <- c(2L, 3, "3")  
w
```

```
[1] "2" "3" "3"
```

```
w <- c(2L, 3L, 3)
w
```

```
[1] 2 3 3
```

```
typeof(w)
```

```
[1] "double"
```

```
z <- c(3, 5, 1, 9)
```

```
z[-1]
```

```
[1] 5 1 9
```

```
z[-3]
```

```
[1] 3 5 9
```

```
z[c(-1, -3)]
```

```
[1] 5 9
```

```
z[-1, -3]
```

```
z[-c(3:2)]
```

```
[1] 3 9
```

```
z[-c(1, 3)]
```

```
[1] 5 9
```

```
x <- 4:31
```

```
head(x)
```

```
[1] 4 5 6 7 8 9
```

```
head(x, n = 3)
```

```
[1] 4 5 6
```

```
head(x, 3)
```

```
[1] 4 5 6
```

```
head(3, x)
```

```
head(n = 3, x = x)
```

```
[1] 4 5 6
```

```
head(n = 3, x = mtcars)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1

```
tail(n = 3, x = mtcars)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
19.7	6	145	175	3.62	2.77	15.5	0	1	5	6
15	8	301	335	3.54	3.57	14.6	0	1	5	8
21.4	4	121	109	4.11	2.78	18.6	1	1	4	2

```
z <- c(3, 5, 1, 9)
```

```
tail(z, n = 2)
```

```
[1] 1 9
```

```
tail(z, n = -2)
```

```
[1] 1 9
```

```
z <- c(z, 7); z
```

```
[1] 3 5 1 9 7
```

```
tail(z, n = 2)
```

```
[1] 9 7
```

Bruk av tail

```
tail(z, n = -2)
```

```
[1] 1 9 7
```

```
x <- rnorm(10)  
q <- x < 0
```

```
typeof(q)
```

```
[1] "logical"
```

```
q
```

```
[1] TRUE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
```

```
q + 5
```

```
[1] 6 5 5 6 5 5 5 5 5
```

```
l <- list(x = 1:3, y = c("A", letters[4:13]), z = seq(from = 1.3, to = 2.7,  
by = 0.2))
```

```
ll <- l
```

```
rm(ll)
```

```
mtcars
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1
21.4	6	258	110	3.08	3.21	19.4	1	0	3	1
18.7	8	360	175	3.15	3.44	17	0	0	3	2
18.1	6	225	105	2.76	3.46	20.2	1	0	3	1
14.3	8	360	245	3.21	3.57	15.8	0	0	3	4
24.4	4	147	62	3.69	3.19	20	1	0	4	2
22.8	4	141	95	3.92	3.15	22.9	1	0	4	2
19.2	6	168	123	3.92	3.44	18.3	1	0	4	4
17.8	6	168	123	3.92	3.44	18.9	1	0	4	4
16.4	8	276	180	3.07	4.07	17.4	0	0	3	3
17.3	8	276	180	3.07	3.73	17.6	0	0	3	3
15.2	8	276	180	3.07	3.78	18	0	0	3	3
10.4	8	472	205	2.93	5.25	18	0	0	3	4
10.4	8	460	215	3	5.42	17.8	0	0	3	4
14.7	8	440	230	3.23	5.34	17.4	0	0	3	4
32.4	4	78.7	66	4.08	2.2	19.5	1	1	4	1
30.4	4	75.7	52	4.93	1.61	18.5	1	1	4	2
33.9	4	71.1	65	4.22	1.83	19.9	1	1	4	1
21.5	4	120	97	3.7	2.46	20	1	0	3	1
15.5	8	318	150	2.76	3.52	16.9	0	0	3	2
15.2	8	304	150	3.15	3.44	17.3	0	0	3	2
13.3	8	350	245	3.73	3.84	15.4	0	0	3	4
19.2	8	400	175	3.08	3.85	17.1	0	0	3	2
27.3	4	79	66	4.08	1.94	18.9	1	1	4	1
26	4	120	91	4.43	2.14	16.7	0	1	5	2
30.4	4	95.1	113	3.77	1.51	16.9	1	1	5	2
15.8	8	351	264	4.22	3.17	14.5	0	1	5	4
19.7	6	145	175	3.62	2.77	15.5	0	1	5	6
15	8	301	335	3.54	3.57	14.6	0	1	5	8
21.4	4	121	109	4.11	2.78	18.6	1	1	4	2

```
typeof(mtcars)
```

```
[1] "list"
```

```
mtcars$mpg
```

```
[1] 21.0 21.0 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 17.8 16.4 17.3 15.2  
10.4  
[16] 10.4 14.7 32.4 30.4 33.9 21.5 15.5 15.2 13.3 19.2 27.3 26.0 30.4 15.8  
19.7  
[31] 15.0 21.4
```

```
typeof(mtcars$mpg)
```

```
[1] "double"
```

Hirarki

1. logical
2. integer
3. numeric
4. character

```
x <- c(TRUE, FALSE, FALSE)
```

```
x
```

```
[1] TRUE FALSE FALSE
```

```
typeof(x)
```

```
[1] "logical"
```

```
x <- c(x, 1L)
```

```
x
```

```
[1] 1 0 0 1
```

```
typeof(x)
```

```
[1] "integer"
```

```
x[5] <- 2.1
```

```
x
```

```
[1] 1.0 0.0 0.0 1.0 2.1
```

```
typeof(x)
```

```
[1] "double"
```

```
x[6] <- "a"
```

```
x
```

```
[1] "1" "0" "0" "1" "2.1" "a"
```

```
df <- data.frame(x = 1:3, y = letters[1:3], z = LETTERS[1:3])
```

```
df[2, c(1, "z")]
```

```
which(row.names(mtcars) == "Toyota Corolla")
```

```
[1] 20
```

```
?which
```

```
head(mtcars)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1
21.4	6	258	110	3.08	3.21	19.4	1	0	3	1
18.7	8	360	175	3.15	3.44	17	0	0	3	2
18.1	6	225	105	2.76	3.46	20.2	1	0	3	1

```
names(mtcars)
```

```
[1] "mpg" "cyl" "disp" "hp" "drat" "wt" "qsec" "vs" "am" "gear"
[11] "carb"
```

```
row.names(mtcars)
```

```
[1] "Mazda RX4"           "Mazda RX4 Wag"       "Datsun 710"
[4] "Hornet 4 Drive"      "Hornet Sportabout"   "Valiant"
[7] "Duster 360"          "Merc 240D"           "Merc 230"
[10] "Merc 280"            "Merc 280C"           "Merc 450SE"
[13] "Merc 450SL"          "Merc 450SLC"         "Cadillac Fleetwood"
[16] "Lincoln Continental" "Chrysler Imperial"   "Fiat 128"
[19] "Honda Civic"         "Toyota Corolla"      "Toyota Corona"
[22] "Dodge Challenger"    "AMC Javelin"         "Camaro Z28"
[25] "Pontiac Firebird"    "Fiat X1-9"           "Porsche 914-2"
[28] "Lotus Europa"        "Ford Pantera L"      "Ferrari Dino"
[31] "Maserati Bora"       "Volvo 142E"
```

```
row.names(mtcars) == "Toyota Corolla"
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
[25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
which(row.names(mtcars) == "Toyota Corolla")
```

```
[1] 20
```

```
mtcars[which(row.names(mtcars) == "Toyota Corolla"),]
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
33.9	4	71.1	65	4.22	1.83	19.9	1	1	4	1

```
mtcars
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1
21.4	6	258	110	3.08	3.21	19.4	1	0	3	1
18.7	8	360	175	3.15	3.44	17	0	0	3	2
18.1	6	225	105	2.76	3.46	20.2	1	0	3	1
14.3	8	360	245	3.21	3.57	15.8	0	0	3	4
24.4	4	147	62	3.69	3.19	20	1	0	4	2
22.8	4	141	95	3.92	3.15	22.9	1	0	4	2
19.2	6	168	123	3.92	3.44	18.3	1	0	4	4
17.8	6	168	123	3.92	3.44	18.9	1	0	4	4
16.4	8	276	180	3.07	4.07	17.4	0	0	3	3
17.3	8	276	180	3.07	3.73	17.6	0	0	3	3
15.2	8	276	180	3.07	3.78	18	0	0	3	3
10.4	8	472	205	2.93	5.25	18	0	0	3	4
10.4	8	460	215	3	5.42	17.8	0	0	3	4
14.7	8	440	230	3.23	5.34	17.4	0	0	3	4
32.4	4	78.7	66	4.08	2.2	19.5	1	1	4	1
30.4	4	75.7	52	4.93	1.61	18.5	1	1	4	2
33.9	4	71.1	65	4.22	1.83	19.9	1	1	4	1
21.5	4	120	97	3.7	2.46	20	1	0	3	1
15.5	8	318	150	2.76	3.52	16.9	0	0	3	2
15.2	8	304	150	3.15	3.44	17.3	0	0	3	2
13.3	8	350	245	3.73	3.84	15.4	0	0	3	4
19.2	8	400	175	3.08	3.85	17.1	0	0	3	2
27.3	4	79	66	4.08	1.94	18.9	1	1	4	1
26	4	120	91	4.43	2.14	16.7	0	1	5	2
30.4	4	95.1	113	3.77	1.51	16.9	1	1	5	2
15.8	8	351	264	4.22	3.17	14.5	0	1	5	4
19.7	6	145	175	3.62	2.77	15.5	0	1	5	6
15	8	301	335	3.54	3.57	14.6	0	1	5	8
21.4	4	121	109	4.11	2.78	18.6	1	1	4	2

```
mtcars[which(row.names(mtcars) == "Lotus Europa"),]
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
30.4	4	95.1	113	3.77	1.51	16.9	1	1	5	2

```
mtcars[which(row.names(mtcars) == "Lotus Europa"), 10]
```

```
[1] 5
```

```
mtcars[which(row.names(mtcars) == "Lotus Europa"), "gear"]
```

```
[1] 5
```

```
modell <- c("Porsche 914-2", "Lotus Europa", "Maserati Bora")
```

```
modell
```

```
[1] "Porsche 914-2" "Lotus Europa" "Maserati Bora"
```

```
row.names(mtcars) %in% modell
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  
[25] FALSE FALSE TRUE TRUE FALSE FALSE TRUE FALSE
```

```
which(row.names(mtcars) %in% modell)
```

```
[1] 27 28 31
```

```
mtcars[which(row.names(mtcars) %in% modell), ]
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
26	4	120	91	4.43	2.14	16.7	0	1	5	2
30.4	4	95.1	113	3.77	1.51	16.9	1	1	5	2
15	8	301	335	3.54	3.57	14.6	0	1	5	8

```
mtcars[which(row.names(mtcars) %in% modell), c("mpg", "hp", "gear")]
```

mpg	hp	gear
26	91	5
30.4	113	5
15	335	5

```
df <- data.frame(x = 1:3, y = letters[1:3], z = LETTERS[1:3])
```

```
df
```

```
df$q <- c(2.1, 0.4, 0.9)
```

Overskrift

```
df
```

x	y	z	q
1	a	A	2.1
2	b	B	0.4
3	c	C	0.9

A friend once said:

It's always better to give than to receive.

```
tib <- as_tibble(df)
```

```
tib
```

x	y	z	q
1	a	A	2.1
2	b	B	0.4
3	c	C	0.9

```
df
```

x	y	z	q
1	a	A	2.1
2	b	B	0.4
3	c	C	0.9

```
is.data.frame(tib)
```

```
[1] TRUE
```

```
is.list(tib)
```

```
[1] TRUE
```

```
is.vector(df[2,3])
```

```
[1] TRUE
```

```
is.data.frame(df[2,3])
```

```
[1] FALSE
```

```
is.vector(tib[2,3])
```

```
[1] FALSE
```

```
is.data.frame(tib[2,3])
```

```
[1] TRUE
```

```
w <- tib[2,3]
```

```
w
```

z
B

```
df[2,3]
```

```
[1] "B"
```

```
# define a matrix in R  
X <- matrix(c(2, 2, -1, 5, 2, 6), nrow = 3, byrow = FALSE)
```

```
Y <- matrix(c(-3, 2, 2.5, 6, -4, 4), nrow = 2, byrow = FALSE)
```

```
X %*% Y
```

```
      [,1] [,2] [,3]  
[1,]     4 35.0  12  
[2,]    -2 17.0   0  
[3,]    15 33.5  28
```

```
Y %*% X
```

```
      [,1] [,2]  
[1,]     3 -34  
[2,]    12  46
```

```
Y
```

```
      [,1] [,2] [,3]  
[1,]    -3  2.5  -4  
[2,]     2  6.0   4
```

```
X
```

```
      [,1] [,2]  
[1,]     2     5  
[2,]     2     2  
[3,]    -1     6
```

```
subset(cars, subset = cars$speed > 6 & cars$speed < 11)
```

speed	dist
7	4
7	22
8	16
9	10
10	18
10	26
10	34

```
cars_sub <- subset(cars, subset = cars$speed > 6 & cars$speed < 11);
cars_sub
```

speed	dist
7	4
7	22
8	16
9	10
10	18
10	26
10	34

```
cars_sub1 <- subset(cars, subset = cars$speed > 6 & cars$speed < 11 &
cars$dist > 15)
```

```
lm1 <- lm(
  formula = dist ~ speed,
  data = cars
)
```

```
summary(lm1)
```

```
Call:
lm(formula = dist ~ speed, data = cars)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-29.069  -9.525  -2.272   9.215  43.201
```

```
Coefficients:
```

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791      6.7584  -2.601   0.0123 *
speed        3.9324      0.4155   9.464 1.49e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared:  0.6511,    Adjusted R-squared:  0.6438
F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12

```

```

lm2 <- lm(
  formula = dist ~ speed,
  data = cars,
  subset = cars$speed > 6 & cars$speed < 11 & cars$dist > 15
)

```

```
summary(lm2)
```

```

Call:
lm(formula = dist ~ speed, data = cars, subset = cars$speed >
    6 & cars$speed < 11 & cars$dist > 15)

Residuals:
    4     5     7     8     9
3.30 -4.95 -7.45  0.55  8.55

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.950      23.780   0.124   0.909
speed          2.250       2.616   0.860   0.453

Residual standard error: 7.4 on 3 degrees of freedom
Multiple R-squared:  0.1978,    Adjusted R-squared:  -0.06966
F-statistic: 0.7395 on 1 and 3 DF,  p-value: 0.4531

```

```

lm3 <- lm(
  formula = dist ~ speed,
  data = cars_sub1
)

```

```
summary(lm3)
```

```

Call:
lm(formula = dist ~ speed, data = cars_sub1)

```

Residuals:

4	5	7	8	9
3.30	-4.95	-7.45	0.55	8.55

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.950	23.780	0.124	0.909
speed	2.250	2.616	0.860	0.453

Residual standard error: 7.4 on 3 degrees of freedom

Multiple R-squared: 0.1978, Adjusted R-squared: -0.06966

F-statistic: 0.7395 on 1 and 3 DF, p-value: 0.4531

Ta vare på ting

- `save()`
- `load()`

```
getwd()
```

```
[1] "/Users/ag/Dev/FirstRepoH26"
```

```
save(cars_sub1, lm1, lm2, file = "my-stuff.RData")
```

```
ls()
```

```
[1] "cars_sub" "cars_sub1" "df"      "l"      "lm1"
"lm2"
[7] "lm3"      "modell"     "q"      "tib"     "w"
"x"
[13] "X"      "Y"      "z"
```

```
rm(cars_sub1, lm1, lm2)
```

```
ls()
```

```
[1] "cars_sub" "df"      "l"      "lm3"      "modell"   "q"
[7] "tib"      "w"      "x"      "X"      "Y"      "z"
```

```
# Merk at navnene er på plass
load(file = "my-stuff.RData")
```

```
ls()
```

```
[1] "cars_sub" "cars_sub1" "df"      "l"      "lm1"
"lm2"
[7] "lm3"      "modell"    "q"       "tib"    "w"
"x"
[13] "X"        "Y"        "z"
```

```
huxreg(
  lm1, lm2, lm3,
  error_format = "{statistic}",
  note         = "{stars}. T statistics in parentheses."
)
```

	(1)	(2)	(3)
(Intercept)	-17.579 *	2.950	2.950
	(-2.601)	(0.124)	(0.124)
speed	3.932 ***	2.250	2.250
	(9.464)	(0.860)	(0.860)
N	50	5	5
R2	0.651	0.198	0.198
logLik	-206.578	-15.825	-15.825
AIC	419.157	37.651	37.651

*** p < 0.001; ** p < 0.01; * p < 0.05. T statistics in parentheses.

```
add_numbers <- function(x, y) {
  x + y
}
```

```
add_numbers(3, 4)
```

```
[1] 7
```

```
add_numbers <- function(x, y) {
  res = x + y
  res
}
```

```
add_numbers(3, 4)
```

```
[1] 7
```

```
add_numbers <- function(x = 0, y = 0) {  
  res = x + y  
  res  
}
```

```
add_numbers()
```

```
[1] 0
```

```
add_numbers(5)
```

```
[1] 5
```

```
df
```

x	y	z	q
1	a	A	2.1
2	b	B	0.4
3	c	C	0.9

```
start_stupid_fun <- function(x1 = 0, n = 10, d = 1) {  
  # initialise x  
  x = vector(mode = "numeric", length = n)  
  x[1] = x1  
  
  for (i in seq_along(x[-1])) {  
    x[i + 1] = x[i] + d  
  }  
  print(x)  
}
```

```
start_stupid_fun()
```

```
[1] 0 1 2 3 4 5 6 7 8 9
```

```
start_stupid_fun(x1 = 17, n = 6)
```

```
[1] 17 18 19 20 21 22
```

```
start_stupid_fun(x1 = 17, n = 6, d = 2)
```

```
[1] 17 19 21 23 25 27
```

```
start_stupid_fun(x1 = 17, n = 6, d = 0.5)
```

```
[1] 17.0 17.5 18.0 18.5 19.0 19.5
```

```
head_and_tail <- function(x) {  
  c(head(x), tail(x))  
}
```

```
head_and_tail(x = mtcars)
```

```
$mpg  
[1] 21.0 21.0 22.8 21.4 18.7 18.1  
  
$cyl  
[1] 6 6 4 6 8 6  
  
$disp  
[1] 160 160 108 258 360 225  
  
$hp  
[1] 110 110 93 110 175 105  
  
$drat  
[1] 3.90 3.90 3.85 3.08 3.15 2.76  
  
$wt  
[1] 2.620 2.875 2.320 3.215 3.440 3.460  
  
$qsec  
[1] 16.46 17.02 18.61 19.44 17.02 20.22  
  
$vs  
[1] 0 0 1 1 0 1  
  
$am  
[1] 1 1 1 0 0 0  
  
$gear  
[1] 4 4 4 3 3 3  
  
$carb  
[1] 4 4 1 1 2 1  
  
$mpg  
[1] 26.0 30.4 15.8 19.7 15.0 21.4
```

```
$cyl
[1] 4 4 8 6 8 4

$disp
[1] 120.3  95.1 351.0 145.0 301.0 121.0

$hp
[1]  91 113 264 175 335 109

$drat
[1] 4.43 3.77 4.22 3.62 3.54 4.11

$wt
[1] 2.140 1.513 3.170 2.770 3.570 2.780

$qsec
[1] 16.7 16.9 14.5 15.5 14.6 18.6

$vs
[1] 0 1 0 0 0 1

$am
[1] 1 1 1 1 1 1

$gear
[1] 5 5 5 5 5 4

$carb
[1] 2 2 4 6 8 2
```

```
mtcars
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1
21.4	6	258	110	3.08	3.21	19.4	1	0	3	1
18.7	8	360	175	3.15	3.44	17	0	0	3	2
18.1	6	225	105	2.76	3.46	20.2	1	0	3	1
14.3	8	360	245	3.21	3.57	15.8	0	0	3	4
24.4	4	147	62	3.69	3.19	20	1	0	4	2
22.8	4	141	95	3.92	3.15	22.9	1	0	4	2
19.2	6	168	123	3.92	3.44	18.3	1	0	4	4
17.8	6	168	123	3.92	3.44	18.9	1	0	4	4
16.4	8	276	180	3.07	4.07	17.4	0	0	3	3
17.3	8	276	180	3.07	3.73	17.6	0	0	3	3
15.2	8	276	180	3.07	3.78	18	0	0	3	3
10.4	8	472	205	2.93	5.25	18	0	0	3	4
10.4	8	460	215	3	5.42	17.8	0	0	3	4
14.7	8	440	230	3.23	5.34	17.4	0	0	3	4
32.4	4	78.7	66	4.08	2.2	19.5	1	1	4	1
30.4	4	75.7	52	4.93	1.61	18.5	1	1	4	2
33.9	4	71.1	65	4.22	1.83	19.9	1	1	4	1
21.5	4	120	97	3.7	2.46	20	1	0	3	1
15.5	8	318	150	2.76	3.52	16.9	0	0	3	2
15.2	8	304	150	3.15	3.44	17.3	0	0	3	2
13.3	8	350	245	3.73	3.84	15.4	0	0	3	4
19.2	8	400	175	3.08	3.85	17.1	0	0	3	2
27.3	4	79	66	4.08	1.94	18.9	1	1	4	1
26	4	120	91	4.43	2.14	16.7	0	1	5	2
30.4	4	95.1	113	3.77	1.51	16.9	1	1	5	2
15.8	8	351	264	4.22	3.17	14.5	0	1	5	4
19.7	6	145	175	3.62	2.77	15.5	0	1	5	6
15	8	301	335	3.54	3.57	14.6	0	1	5	8
21.4	4	121	109	4.11	2.78	18.6	1	1	4	2

```
head_and_tail <- function(x, n=3L) {
  n_vars = length(names(x))
  res = rbind(
    rep('Head', n_vars),
    head(x, n = n),
    rep('*_*_*', n_vars),
    rep('Tail', n_vars),
    tail(x, n = n)
  )
  as_tibble(res)
}
```

```
head_and_tail(x = mtcars, n = 2)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Head	Head	Head	Head	Head	Head	Head	Head	Head	Head	Head
21	6	160	110	3.9	2.62	16.46	0	1	4	4
21	6	160	110	3.9	2.875	17.02	0	1	4	4
*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*	*_*_*
Tail	Tail	Tail	Tail	Tail	Tail	Tail	Tail	Tail	Tail	Tail
15	8	301	335	3.54	3.57	14.6	0	1	5	8
21.4	4	121	109	4.11	2.78	18.6	1	1	4	2

```
head_and_tail(letters, n = 2)
```

Warning: The `x` argument of `as\_tibble.matrix()` must have unique column names if
 `.name\_repair` is omitted as of tibble 2.0.0.
 i Using compatibility `.name\_repair`.

V1	V2
a	b
y	z